Student Risk Prediction Project

Technical Interview Preparation Guide

Project elevator pitch: I built an end-to-end Python pipeline that cleans student academic data, creates transparent Low/Medium/High risk labels, trains a multi-class softmax logistic-regression classifier, evaluates it with classification metrics, and generates student-specific recommendations and a CSV report.

1. Project at a glance

• **Input:** 10,030 student records with attendance, academic scores, demographics, department, and preparation-course information.

• **Target:** a custom proxy risk label because the source data has no observed intervention/failure outcome.

• **Model:** multi-class softmax logistic regression implemented with NumPy.

• **Outputs:** evaluation metrics JSON, attendance-vs-score visualization, and Student/Attendance/Risk/Recommendation CSV.

• **Execution:** `python main.py`.

2. Architecture and data flow

The system is intentionally a small batch ML pipeline rather than a web service. The entry point calls `run_pipeline()`, which performs the following sequence:

Stage	What happens	Why it exists
1. Load	Read the Kaggle CSV with pandas.	Create a reproducible data source.
2. Clean	Convert numeric fields; replace missing and impossible values with medians.	Prevent invalid values from distorting training.
3. Label	Compute average of Total, Midterm, and Final scores; combine it with attendance to create a proxy risk label.	Create an explainable target where none exists.
4. Features	Select academic/attendance features; one-hot encode Gender and Department.	Convert mixed tabular data into a numeric matrix.
5. Split	Use a seeded stratified 80/20 train/test split.	Preserve class proportions and make results repeatable.
6. Scale	Fit means/std-devs on training data only.	Avoid test-set leakage and improve optimization.
7. Train	Optimize softmax cross-entropy with gradient descent.	Learn probabilities for three risk classes.
8. Evaluate	Calculate accuracy, per-class precision/recall/F1, and confusion matrix.	Measure errors by class, not just overall correctness.
9. Deliver	Refit on all rows after holdout evaluation; export predictions, recommendations, metrics, and visualization artifacts.	Create the requested submission artifacts.

3. Key design decisions: how and why

Why did you choose logistic regression?

The target has three discrete classes and the dataset is tabular, so multi-class logistic regression is an appropriate baseline. It is fast, interpretable, produces class probabilities, and is less likely to overfit than an unconstrained tree on a small/medium feature set. I implemented softmax regression with NumPy because the environment blocked a SciPy binary dependency used by scikit-learn. In a normal production environment I would also benchmark a library implementation.

Why not Random Forest or Decision Tree?

They are valid alternatives and may capture non-linear interactions better. A single tree can overfit and be unstable; Random Forest is stronger but less transparent and requires more tuning. I would compare them using the same split and macro F1, with special attention to High-risk recall. The chosen model is a clear, reliable baseline, not a claim that it is universally optimal.

Why define your own risk labels?

The dataset does not contain a real risk or intervention-outcome column. Without a target, supervised learning cannot learn future risk. I therefore defined an explicit proxy: High if attendance is below 60% or average academic score is below 55; Medium if attendance is below 80% or average score below 75; otherwise Low. These thresholds are policy choices and must be validated with educators or historical outcomes.

Is the model really learning, or just reproducing the rules?

It is learning to reproduce the rule-generated labels from the available features. That is the correct limitation to state. The metrics demonstrate consistency with the proxy criteria, not real-world prediction of future failure. To claim predictive risk, I would need time-ordered historical labels such as course failure, withdrawal, or whether intervention was required.

Why use median imputation?

Median imputation is robust to outliers and keeps every row available for training. It is a simple baseline for numeric tabular data. In a production pipeline, imputation statistics must be fitted on training data only and persisted, and missingness indicator features could be added if missingness itself carries information.

Why one-hot encode categories?

Gender and Department are categorical, not ordinal. One-hot encoding represents each category without falsely implying that, for example, one department is numerically greater than another.

Why standardize features?

Gradient descent is sensitive to feature scale. Attendance and scores are on different ranges, so standardization makes updates more balanced and improves convergence. The mean and standard deviation are calculated from training rows and applied to the test rows, which prevents data leakage.

Why use a stratified split and a fixed seed?

Risk classes are imbalanced, especially Medium. Stratification ensures each split contains all classes in similar proportions. A fixed seed makes the result reproducible for debugging and interview demonstration.

Why not include Student_ID, name, or email?

Those are identifiers, not meaningful predictive signals. Including them can create memorization, privacy risk, and poor generalization. Student_ID is retained only for reporting the final recommendation.

4. Evaluation and interpretation

The latest verified run produced approximately 0.8621 accuracy, 0.8605 macro precision, 0.7938 macro recall, and 0.8221 macro F1. The confusion matrix was:

Actual \ Predicted	Low	Medium	High
Low	190	99	0
Medium	36	1187	48
High	0	94	354

What is accuracy?

The proportion of all predictions that are correct. It is easy to understand but can hide weak performance on minority classes when the data is imbalanced.

What is precision?

For a class, precision is the fraction of predicted members that truly belong to that class. High-risk precision answers: when the system says High, how often is it High?

What is recall and why is High-risk recall important?

Recall is the fraction of actual class members found by the model. High-risk recall is important because a false negative can leave a student without timely support. Depending on the school objective, I would tune the decision threshold to prioritize High-risk recall.

What is F1-score?

F1 is the harmonic mean of precision and recall. It is useful when both false positives and false negatives matter and gives a more balanced view than accuracy alone.

How would you improve the evaluation?

Use repeated stratified cross-validation, compare multiple models, report confidence intervals, inspect class-specific errors, and evaluate on a future time-based holdout. Most importantly, replace proxy labels with verified outcomes before making deployment claims.

The attendance-to-score correlation is about -0.0104. What does that mean?

It indicates almost no linear relationship between those two columns in this dataset. That is a finding, not a failure. Correlation only captures linear association and does not prove causation. Risk labels also use attendance thresholds, so attendance can still be operationally important even when its overall linear correlation is weak.

Why can a student with high attendance still be Medium risk?

The rule uses attendance and academic performance. A student with good attendance but weak average scores can still be Medium or High. This avoids treating attendance as the only signal.

5. Likely technical interview questions

Explain softmax regression mathematically.

For input vector x, each class gets a logit z_k = w_k^T x + b_k. Softmax converts logits to probabilities: p_k = exp(z_k) / sum_j exp(z_j). Training minimizes cross-entropy loss. The gradient is computed from predicted probabilities minus one-hot labels, then weights and biases are updated using gradient descent.

Why subtract the maximum logit before exponentiating?

It is a numerical-stability trick. Subtracting the row maximum does not change the softmax probabilities, but prevents overflow when exponentials receive large values.

What is data leakage?

Leakage occurs when information unavailable at prediction time influences training or preprocessing. Examples include scaling with the complete dataset, using future grades, or putting identifiers into features. The pipeline fits scaling using training rows only and excludes identifiers.

Is there target leakage in this project?

The target is explicitly derived from Total_Score, Midterm_Score, Final_Score, and attendance, while those same fields are also model features. Therefore, the model can reconstruct the proxy rule easily. That is acceptable for demonstrating the requested workflow but is a limitation, not evidence of real predictive power. A better study would predict a future outcome using only earlier-semester features.

How would you redesign the target to avoid leakage?

Define a future target, such as whether a student fails the next term. Use only information available before that term: prior attendance, prior assessments, and preparation history. Split by time, fit preprocessing on past data, and never use future scores as inputs.

How would you deploy this system?

For a first version, schedule the batch script after each grading period, store the predictions in a database, and expose a dashboard to authorized advisors. A REST API could serve one-student predictions, but batch processing is simpler and better aligned with the available CSV workflow.

What would you monitor in production?

Input schema changes, missingness, feature drift, class distribution, prediction distribution, High-risk recall, false-negative review outcomes, latency, failed jobs, and fairness metrics across relevant groups. Retraining should be triggered by validated performance or drift, not by accuracy alone.

How would you handle privacy?

Minimize personally identifiable information, avoid using names and emails as features, restrict advisor access, encrypt data at rest and in transit, audit access, define retention rules, and provide a human review process. Predictions should support—not automatically punish—students.

How would you address fairness?

Compare false-positive and false-negative rates across protected or relevant groups, inspect whether missing data is group-dependent, avoid sensitive attributes unless there is a justified governance reason, and involve educators in defining acceptable trade-offs. Do not assume overall accuracy implies fairness.

What happens if a new department appears?

A production transformer must preserve the training feature schema. Unknown categories should map to an all-zero or explicit unknown category, and schema validation should fail loudly if required numeric columns are missing.

How would you test the pipeline?

Unit-test cleaning bounds, risk-threshold boundaries, one-hot feature consistency, softmax probability sums, confusion-matrix counts, and recommendation text. Add an integration test that runs a small fixture end to end and verifies the three output files.

6. Alternatives and when to choose them

Option	Strength	Weakness	When I would choose it
Logistic regression	Fast, interpretable, probabilistic baseline	Linear decision boundaries	First baseline for tabular classification
Decision tree	Readable non-linear rules	Can overfit and be unstable	Small dataset where explanations must look like rules
Random forest	Strong non-linear baseline and robust	Less transparent; larger model	After baseline when interactions matter
Gradient boosting	Often excellent tabular accuracy	More tuning and explainability work	Production accuracy-focused model with validation
Neural network	Flexible and powerful	Needs more data/tuning; less interpretable	Large-scale or complex unstructured data
Rule-only system	Fully transparent; no training	No learned generalization	Policy prototype or when no labels exist

7. Honest limitations to mention proactively

• The risk target is a proxy created from the same academic variables used as features.

• There is no historical outcome proving that a student later failed or needed intervention.

• The attendance-to-total-score correlation is nearly zero in this dataset.

• The project is a local batch pipeline, not a deployed authenticated service.

• Median imputation is a baseline; production preprocessing should be fitted and persisted as a versioned artifact.

• Model metrics are not enough: threshold selection, fairness, calibration, drift, and human review also matter.

8. Strong closing answer

“I chose a transparent, reproducible baseline because the dataset is tabular and the task asks for explainable risk categories. I cleaned invalid values, encoded categorical features, used a stratified split, prevented scaling leakage, trained softmax logistic regression, and reported class-aware metrics. The most important limitation is that the dataset has no real future-risk label, so my labels are a documented proxy. If I took this further, I would collect time-based intervention or failure outcomes, remove target leakage by using only earlier information, compare tree-based models, calibrate risk probabilities, and validate fairness with educators before deployment.””

Before the interview

• Run `python main.py` once and know where each output is saved.

• Be able to draw the nine-stage pipeline from memory.

• Memorize the risk thresholds and explain why they are only proxy labels.

• Know the difference between precision, recall, F1, and accuracy.

• Say the target-leakage limitation before the interviewer has to discover it.